Outset · AI in Healthcare · Day 3 · Group 5

Predicting heart disease from a checkup

Can a model read 13 routine checkup numbers and flag who is at risk for heart disease -- fairly, for women and men alike?

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Outset

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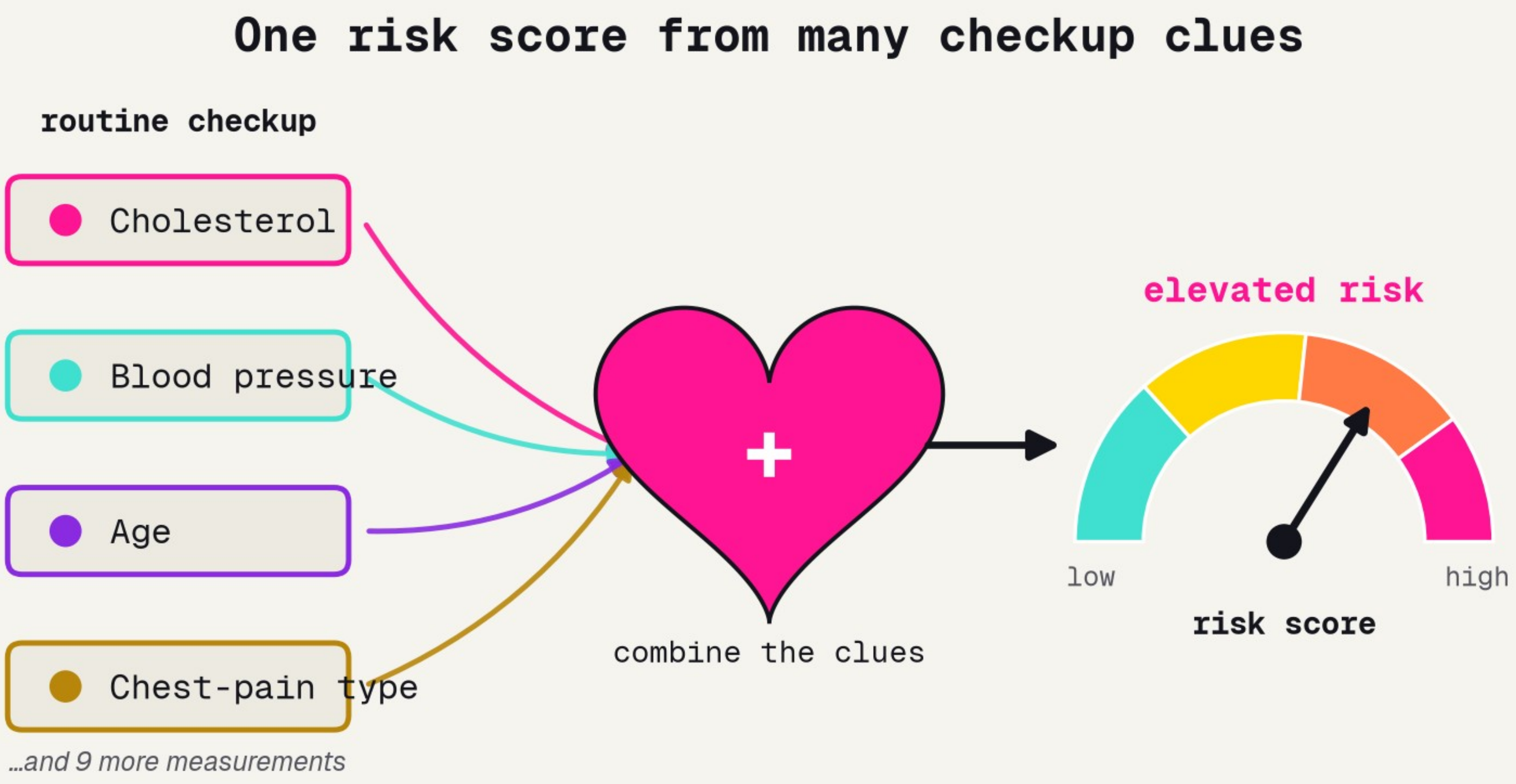
Overview

BACKGROUND

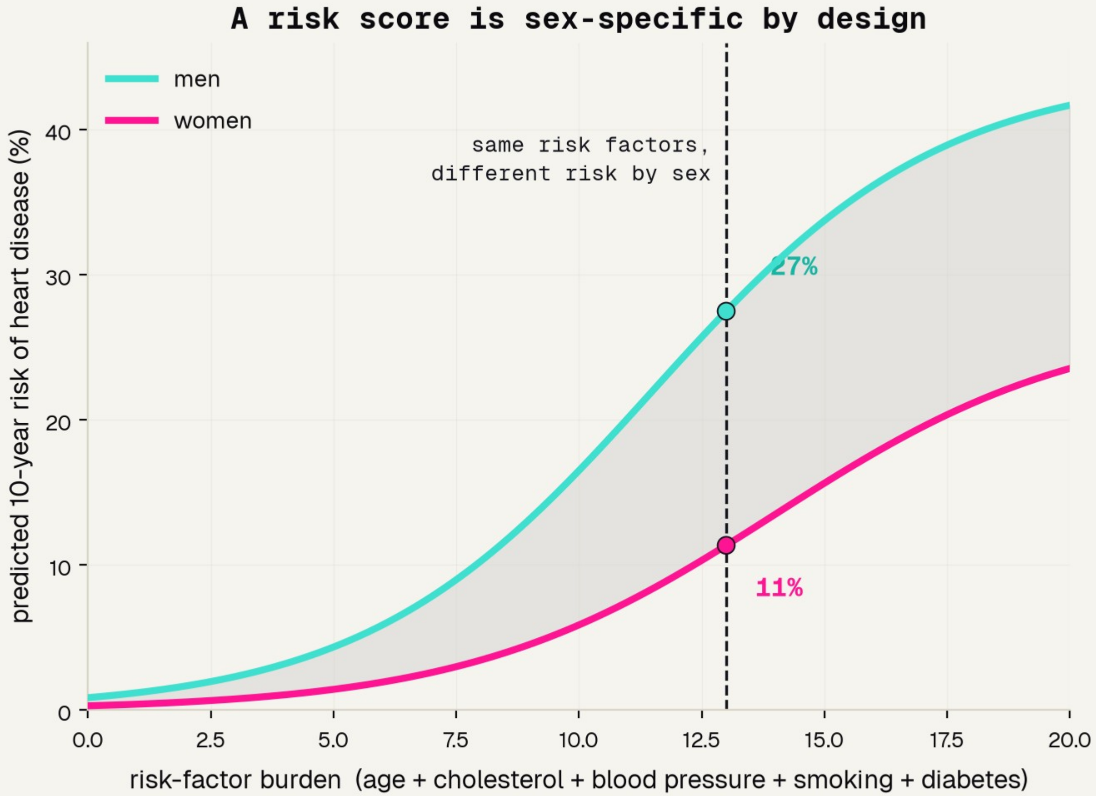
One risk score from many clues



the job a cardiologist's risk calculator does (concept, cf. Framingham -- Wilson 1998)

The checkup measures many clues; a good risk tool combines them into one number -- never trusting a single measurement alone.

Doctors already do this math



adapted from Wilson et al. 1998 (Framingham Risk Score), Circulation

1998

the year Framingham turned checkup clues into a 10-year risk

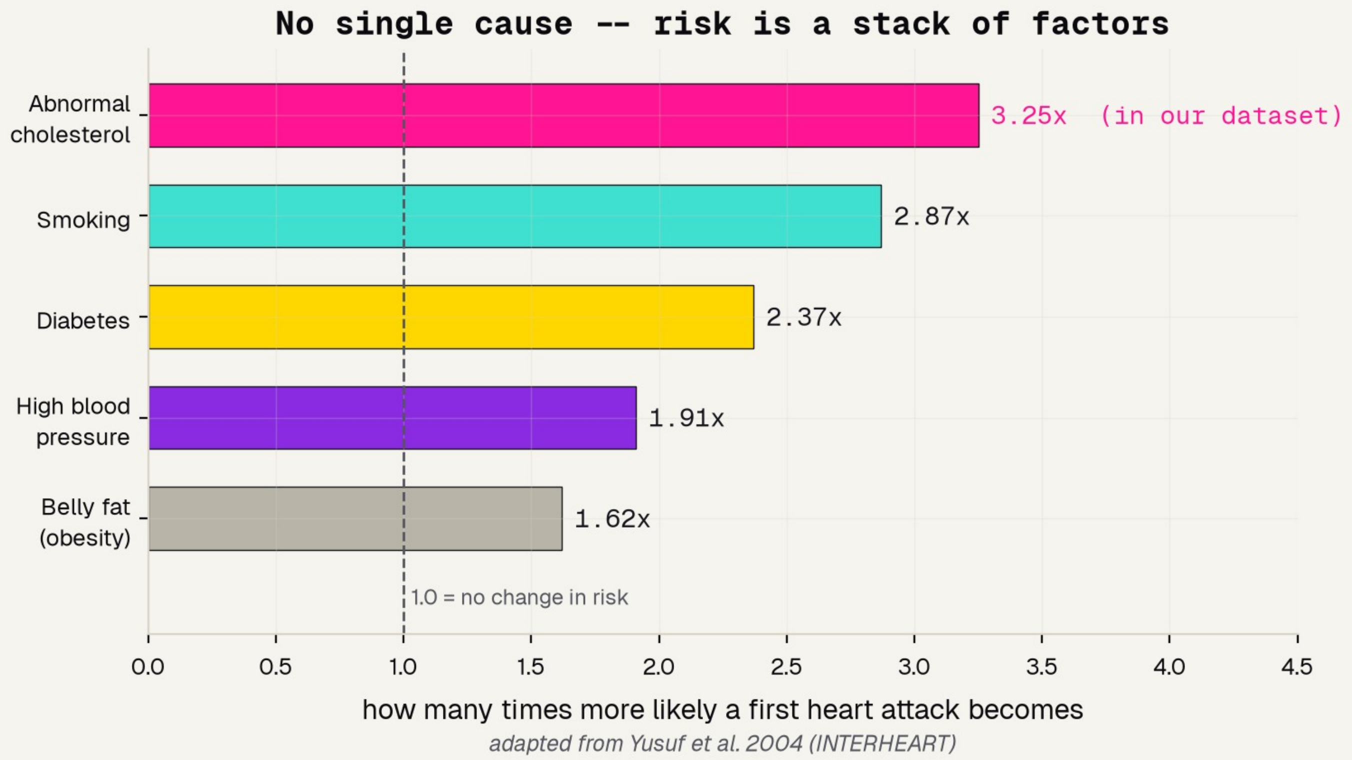
10-yr

the horizon it predicts -- not just today, but the decade ahead

by sex

men and women get different math, because the same numbers mean different things

No single cause: risk is a stack



RISK IS A STACK

No single cause

Nine factors together explain over 90% of first heart attacks.

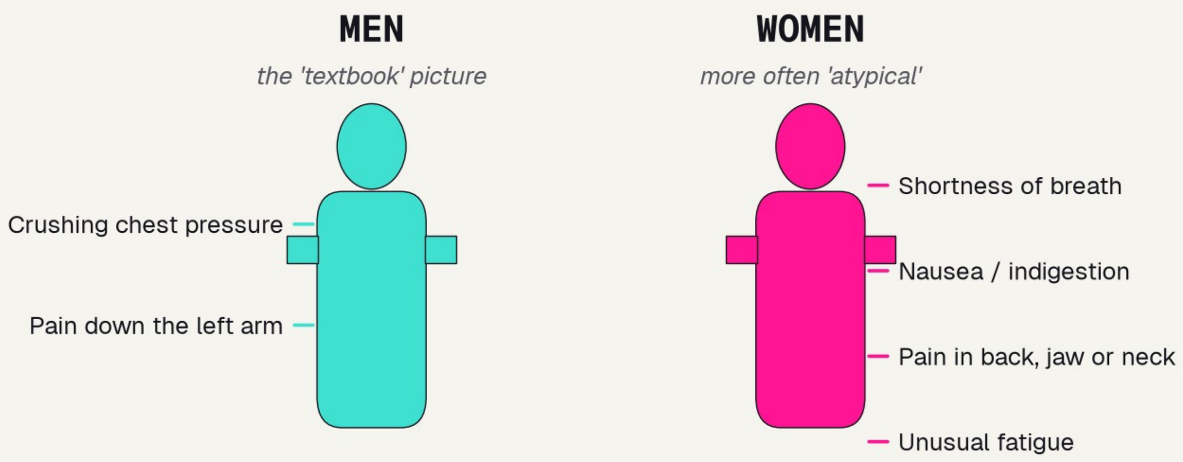
Cholesterol leads

Abnormal cholesterol sits near the top -- and it is one of our 13 features.

So we measure many

A checkup captures several clues at once, never just one number.

The textbook heart attack is the male one



The 'classic' heart-attack symptoms are the MALE ones -- one reason women's heart attacks are more often missed.

concept from an AHA scientific statement (Mehta et al. 2016) and van Oosterhout et al. 2020

THE TEXTBOOK IS MALE

Men more often get the classic crushing chest pain. Women more often get back pain, nausea, or shortness of breath.

Because the classic picture is the male one, women's heart attacks are more often missed.

Watch this thread -- it returns in our fairness check.

Methodology

THE DATA

Why the Cleveland dataset

303

real patient records from a 1980s coronary-disease study, each with a verified yes/no answer

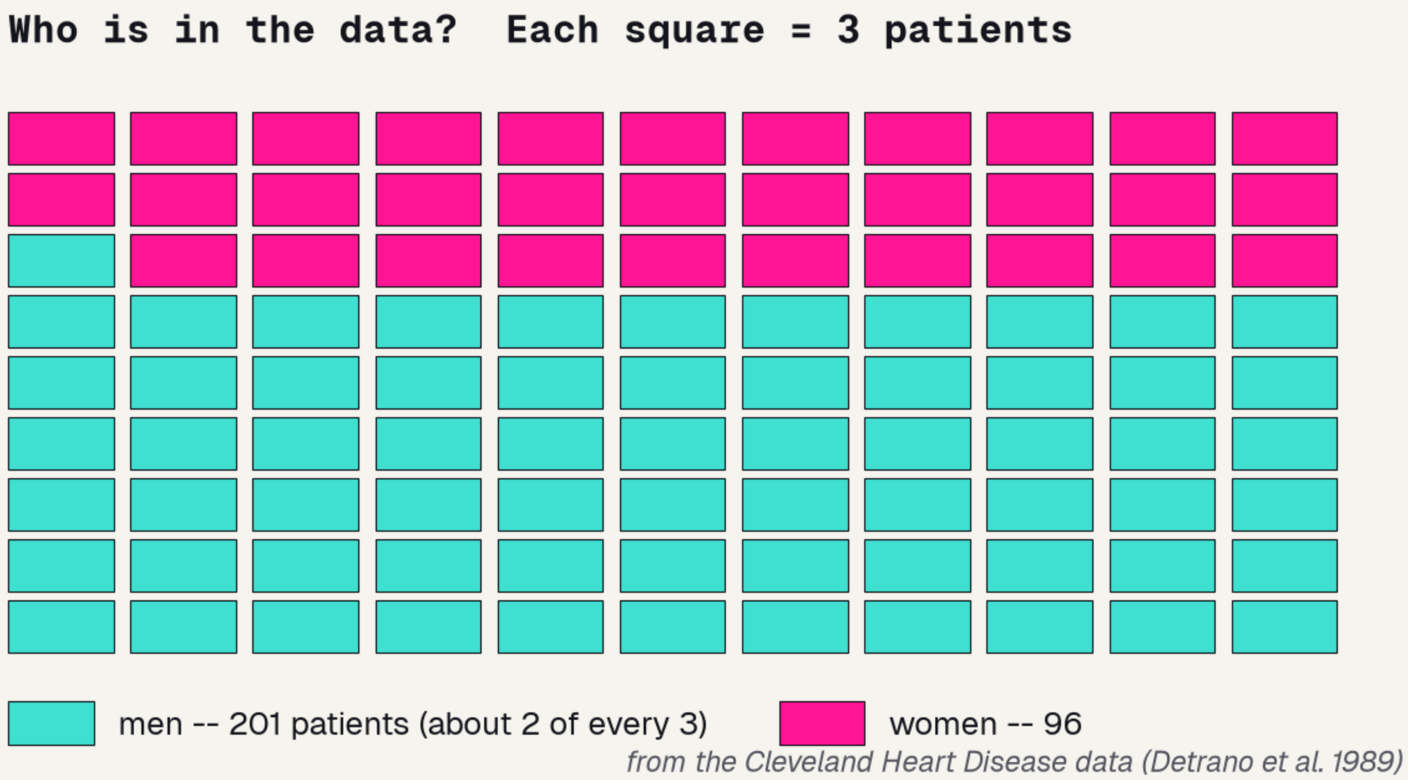
13

routine checkup numbers -- age, blood pressure, cholesterol, chest-pain type, exercise-test results

free

downloads straight from the UCI repository with no gate -- and it records sex, so we can audit fairness

Who is in the data?



2 : 1

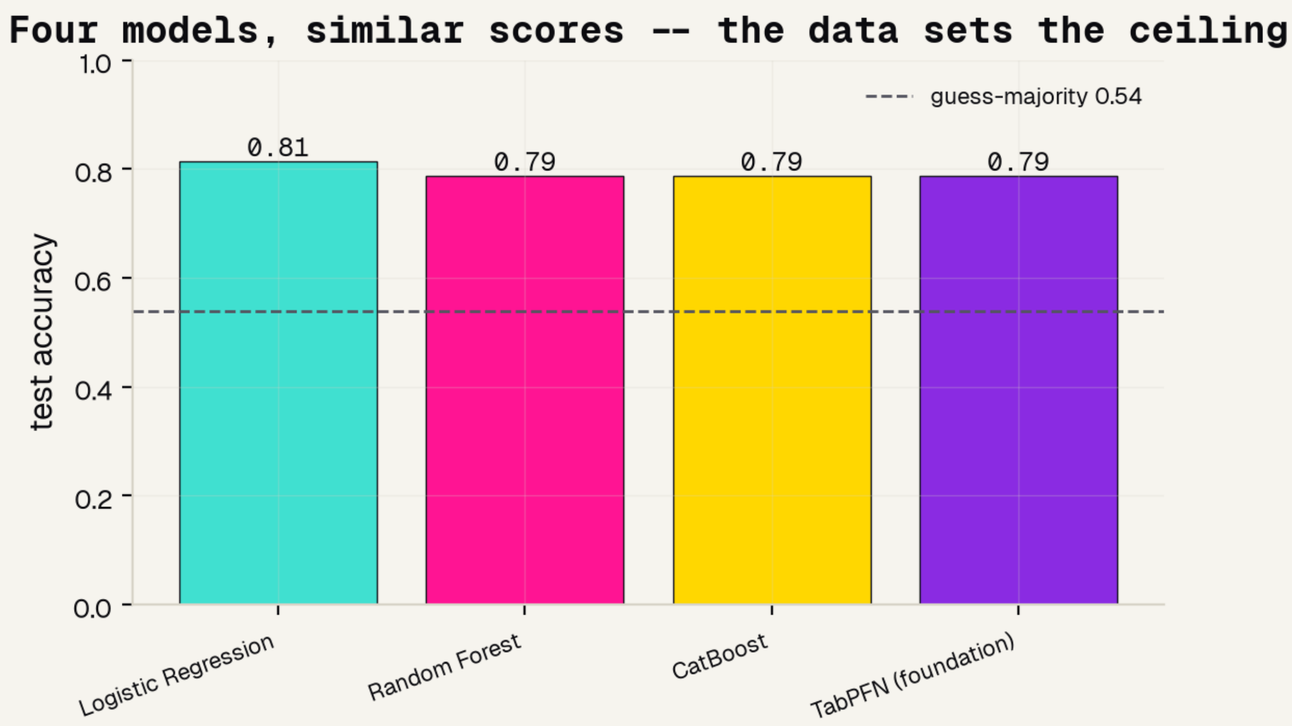
men to women -- the model gets far more practice on men

the honest catch

a model only learns from who is in its data -- this skew returns in our fairness check

THE MODEL

Four models, one bake-off



WE TESTED, WE DID NOT GUESS

~0.80

test accuracy -- and all four models land within a whisker of it.

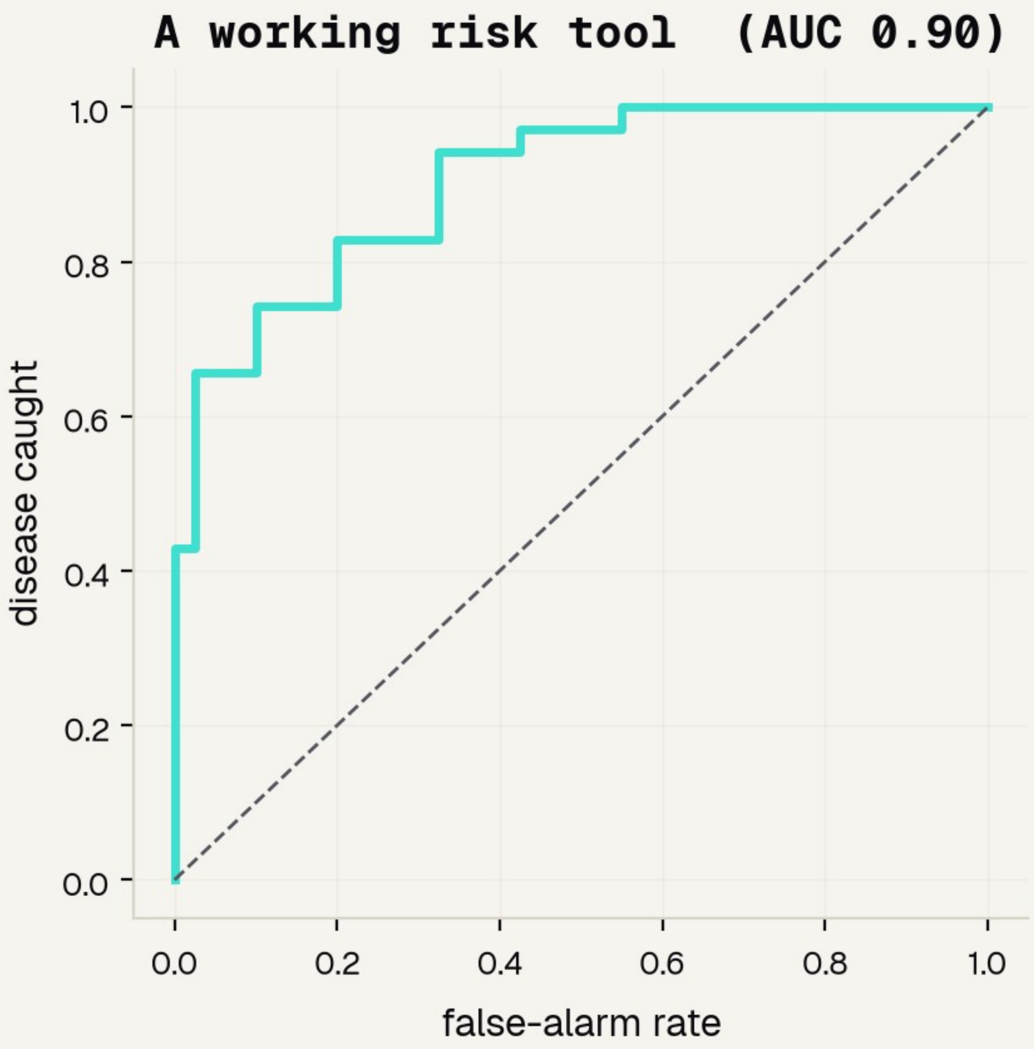
On clean tabular data the model barely matters. The data sets the ceiling -- 300 patients hold only so much signal.

Model and data processing

1	SPLIT	one clean 75/25 train-test split, fixed seed, stratified so both halves carry the same disease rate
2	FEATURES	all 13 checkup measurements, each read as a plain number -- no hand-picking
3	MODEL	CatBoost for the headline score (a calibrated probability we grade by AUC); logistic regression for the clean feature tests
4	GRADE	every number reported on the held-out test set the model never trained on -- no leakage

RESULTS

A working risk tool



0.90

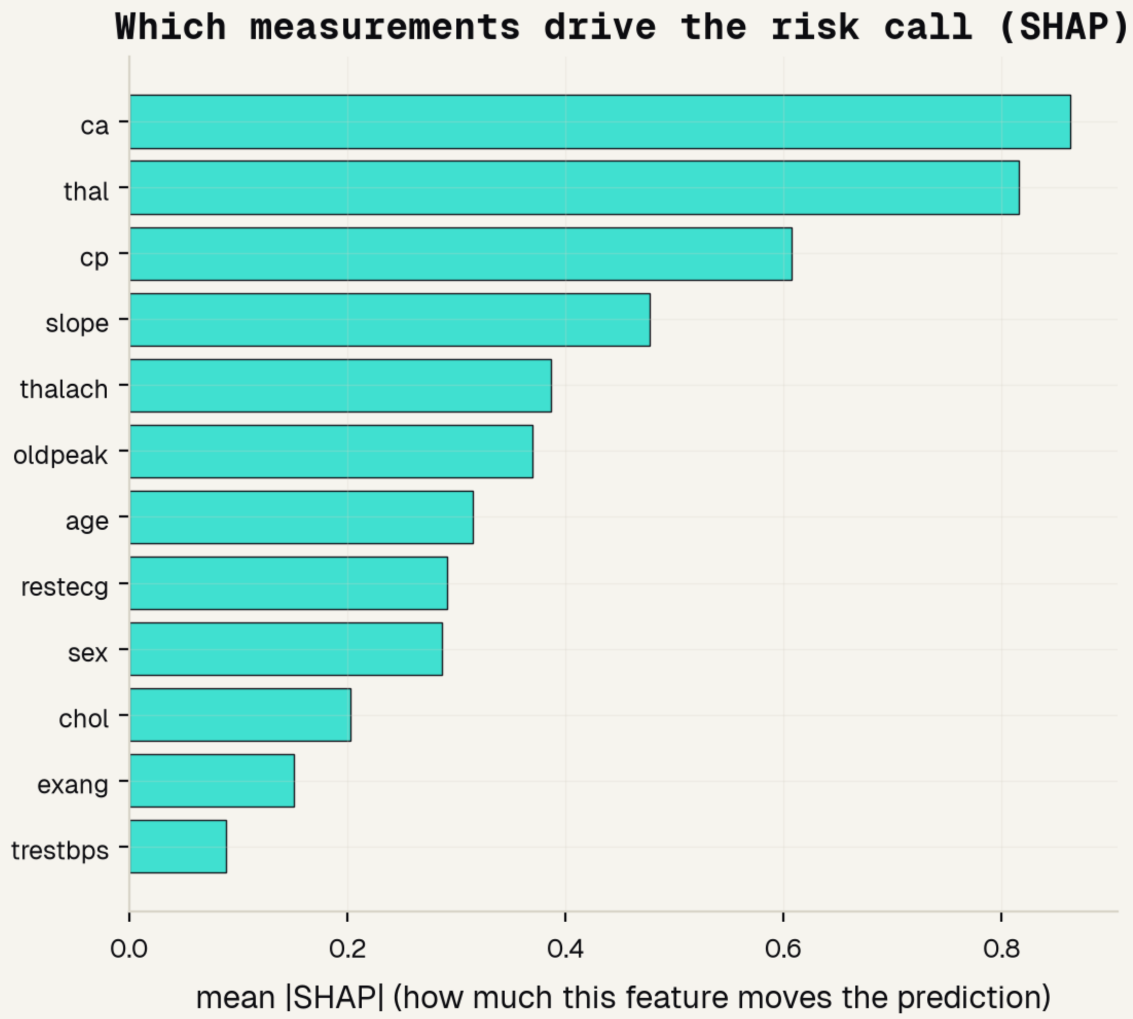
AUC -- the area under the ROC curve

The ROC curve sweeps every cut-off and plots disease caught against false alarms. AUC is the area underneath.

0.5 = a coin flip **1.0 = perfect**

Plain accuracy was only 0.81 -- AUC credits the tool for ranking the sickest patients first.

Which clues drive the call



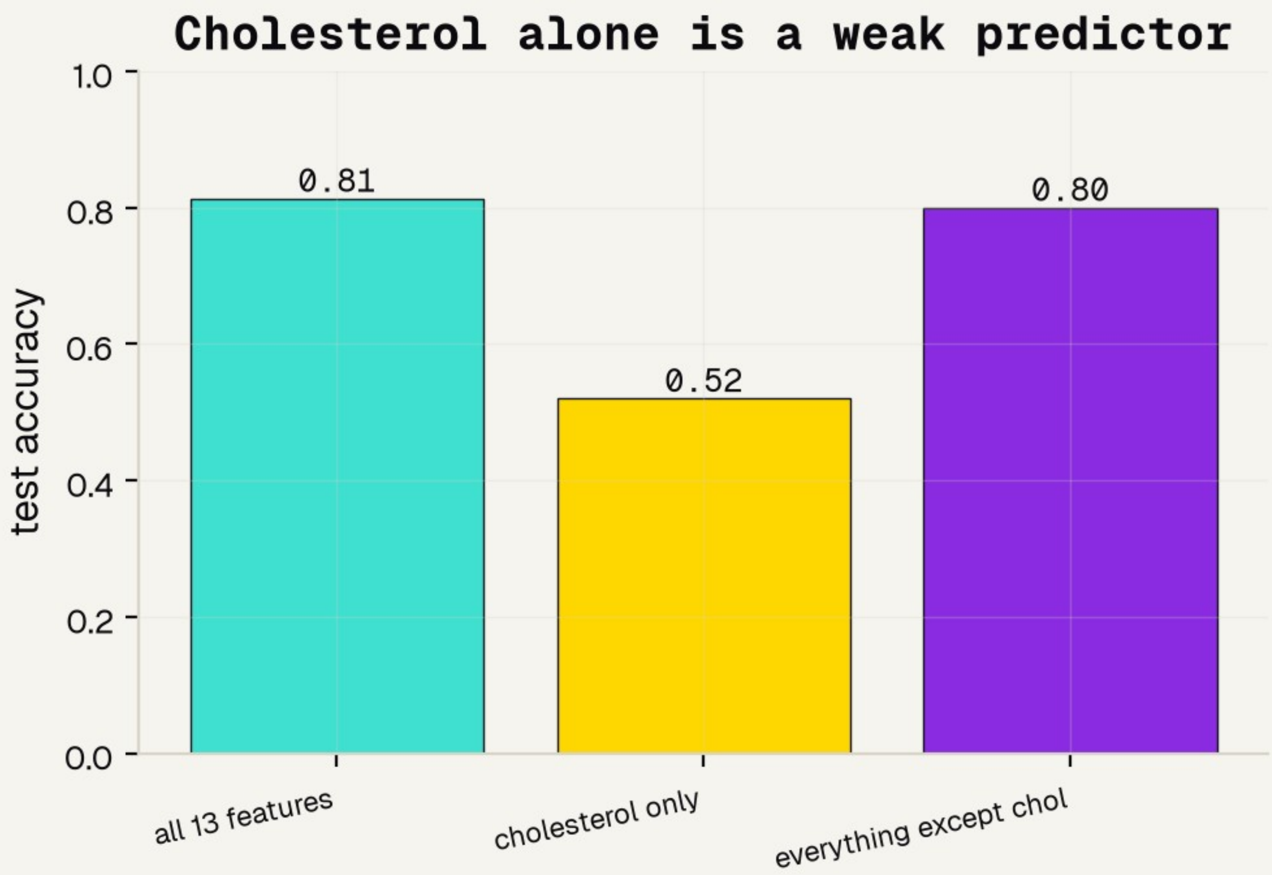
ca vessels seen on imaging -- the top signal

thal a heart stress-test result

cp chest-pain type -- the cue a cardiologist weighs

And cholesterol (chol) sits near the bottom -- a real surprise we test next.

Cholesterol alone is a weak clue



0.52

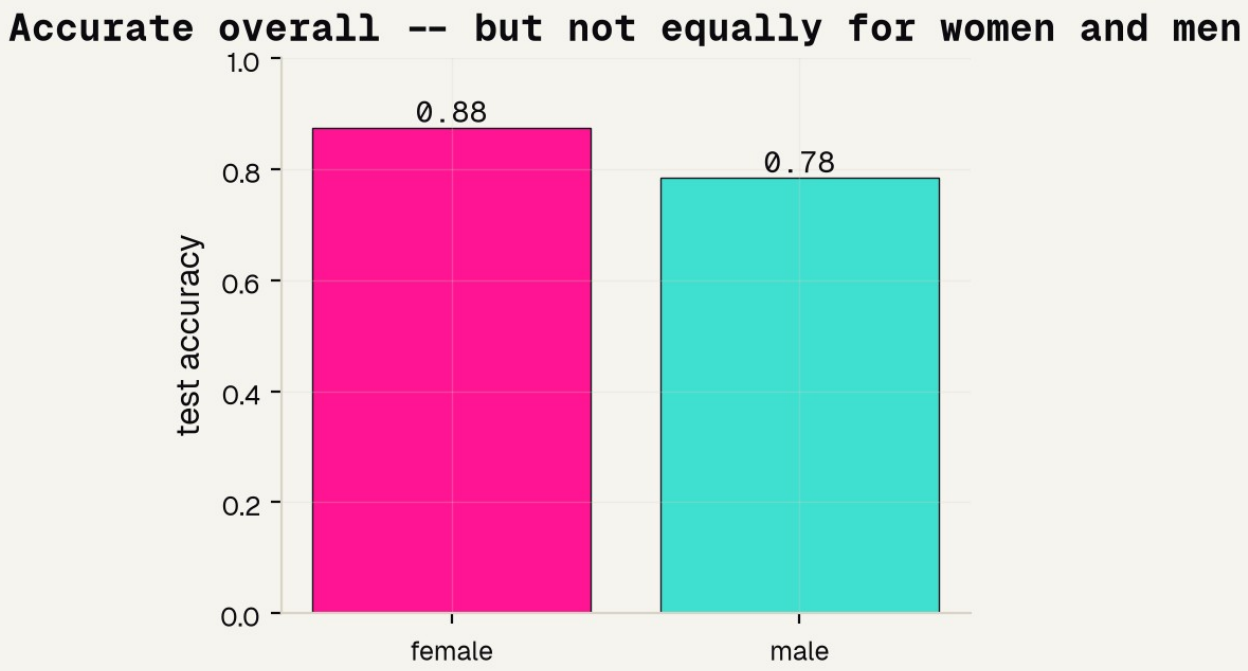
cholesterol ALONE -- barely better than a coin flip

0.80

the checkup WITHOUT cholesterol -- still strong

Causing disease slowly is not the same as detecting it today.

Does it work for women and men?



0.88

accuracy for women

0.78

accuracy for men

A 0.09 gap -- and the honest cause is the 2:1 male data. One overall number would have hidden it.

Overview

THE HONEST PICTURE

What it can and cannot do

IT WORKS

A real risk tool on real checkup data: AUC 0.90, leaning on the same cues (ca, thal, chest-pain type) a cardiologist uses.

AUC 0.90

BUT IT IS SKEWED

The accuracy gap by sex is real, and the data is about 2:1 male -- so it may serve women worse in the real world.

0.09 gap by sex

A REAL TOOL

Would use today's diverse patients, a sex-specific design, multi-hospital validation, and a clinician in the loop.

not ready for a patient

References

FOUNDATIONS & DATA

[1] Detrano et al. 1989, Am J Cardiol -- the Cleveland dataset we use.

[2] Wilson et al. 1998, Circulation -- the Framingham Risk Score.

[3] Yusuf et al. 2004, Lancet -- INTERHEART: nine factors, 90% of risk.

[7] UCI Machine Learning Repository -- Heart Disease Data Set.

SEX DIFFERENCES & UNDER-DIAGNOSIS

[4] Mehta et al. 2016, Circulation -- AHA statement on heart attacks in women.

[5] van Oosterhout et al. 2020, JAHA -- sex differences in symptoms.

[6] Bugiardini & Bairey Merz 2005, JAMA -- angina with "normal" arteries.

The honest bottom line

Grade with the right metric. Ask which clues matter. Check who it works for.

That is how a risk tool earns trust. Use this to learn the judgment -- then use a validated, sex-specific tool and a real clinician to actually care for a patient.

Thank You

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夢想